

Mark "Xavier" Bonavita

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Education

Worcester Polytechnic Institute (WPI) | [Worcester, MA](#)

- **Masters of Science** in Computer Science
- **Bachelor of Science** in Computer Science

GPA 3.62 / 4.0
08/2025 - 05/2026
08/2022 - 05/2026

Technical Skills

Languages

C/C++, Go, Java, Kotlin, JS/TS, JSON, Python, SQL, HTML/CSS

Tools/Frameworks

AWS, Azure, Vue, Docker, REST, Unix, Flask, SQLite, PostgreSQL, Git, React

Professional Experience

Field-Deployable DNA Classification System (NSF-Sponsored) | [Worcester, MA](#)

Mar - Oct 2025

Lead Machine Learning Engineer & UX Consultant

- In a National Science Foundation (NSF) sponsored project, created an AI species identification pipeline using melting curve analysis and a Vision Transformer (ViT) model to classify DNA denaturation patterns
- Achieved 95.35% accuracy and <0.02 loss by applying Optuna hyperparameter tuning and generating synthetic data to resolve class imbalance within our PyTorch model.
- Built a cross-platform React Native app that enables real-time field species identification through a remote classification server.
- Co-authored a research publication demonstrating applications in conservation biology, biodiversity protection, and anti-wildlife trafficking initiatives.

Projects

Recruit.me, Full-Stack Job Matching & Hiring Platform | [Worcester, MA](#)

Aug - Dec 2025

- Built a full-stack recruitment platform using React, React Router, and a RESTful API backed by AWS Lambda functions, enabling responsive, client-side routed UIs and scalable serverless backend operations.
- Designed and implemented the application's backend on AWS RDS (MySQL), developing normalized schemas and efficient query patterns to support complex job, applicant, and company workflows.
- Integrated serverless AWS Lambda endpoints wrapped in a REST API to handle job listings, applicant status changes, company actions, and admin reporting, ensuring reliable, low-latency communication with the frontend.
- Created end-to-end user flows (company/applicant/admin actions), added scalable pagination for large datasets, and produced wireframes, UI diagrams, and detailed requirements documentation through iterative architectural reviews and peer collaboration.

furspace.blog, Humanities Practicum | [Worcester, MA](#)

Jul '24 - Mar 2025

- Alongside two other artists, designed and implemented a nostalgic early-2000s visual storytelling website, drawing inspiration from platforms like MySpace to evoke a distinct digital retro atmosphere
- Wrote a narrative-driven experience, integrating a continuous storyline throughout the blog-style layout to enhance visitor engagement and immersion.
- Developed custom scripts to automate the display of multimedia content and blog entries, ensuring consistent formatting.
- Contributed to the creative direction and visual identity, balancing unique artistic styles within a cohesive design framework.

When2Goat, Meeting and Schedule Coordination Website | [Worcester, MA](#)

Mar - May 2024

- Worked in a team of four to develop a full-stack application within 5 weeks, following Agile/Scrum methodologies, designing and implementing a scalable Flask (Python) backend with an Azure-hosted PostgreSQL database.
- Streamlined user experience (UX) by delivering a simple yet feature-rich interface that functions efficiently across various device types and screen sizes.
- Integrated WPI's Microsoft Single Sign-On (SSO) and Microsoft Entra API for secure identity verification and seamless Microsoft Calendar integration.